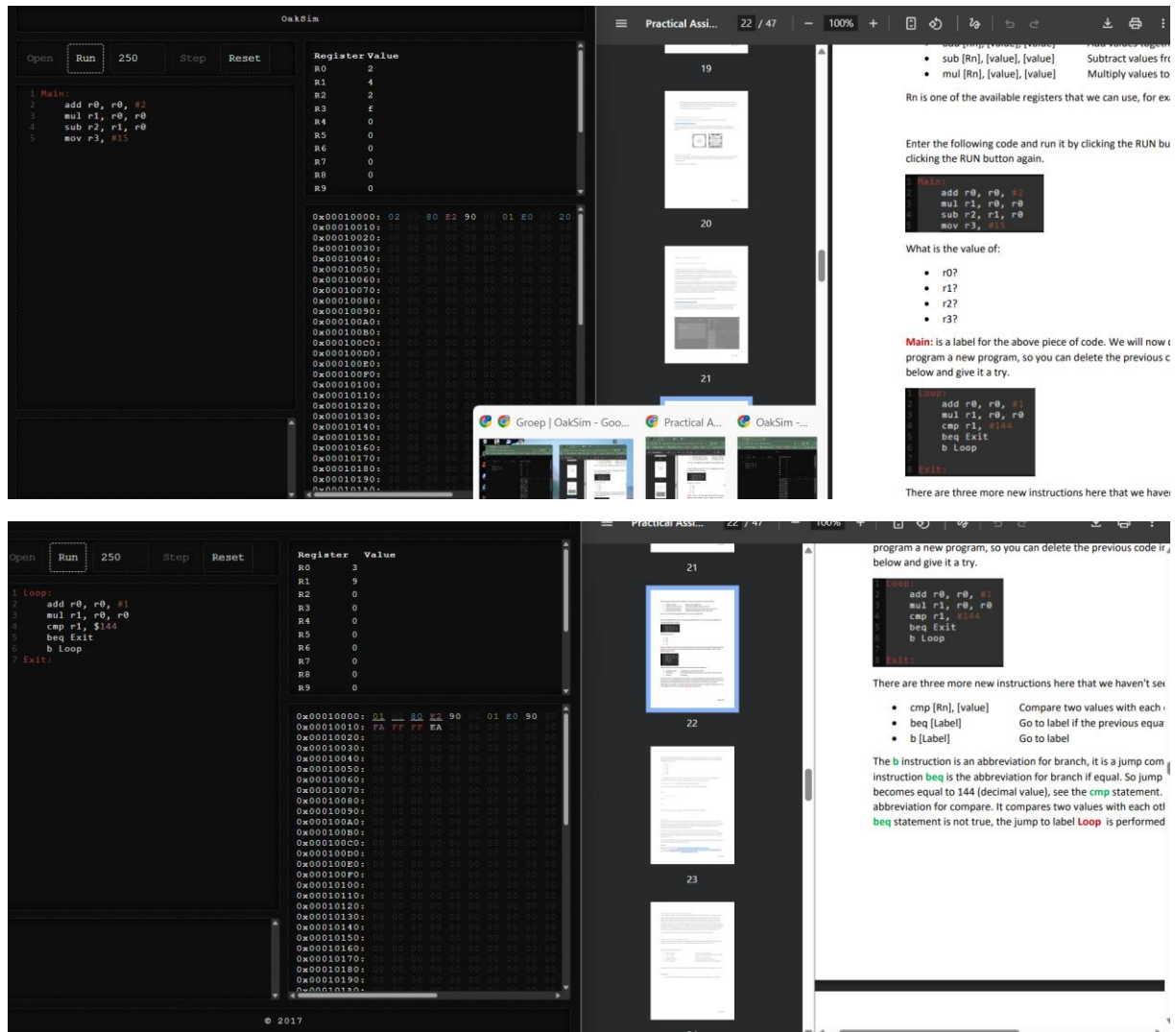


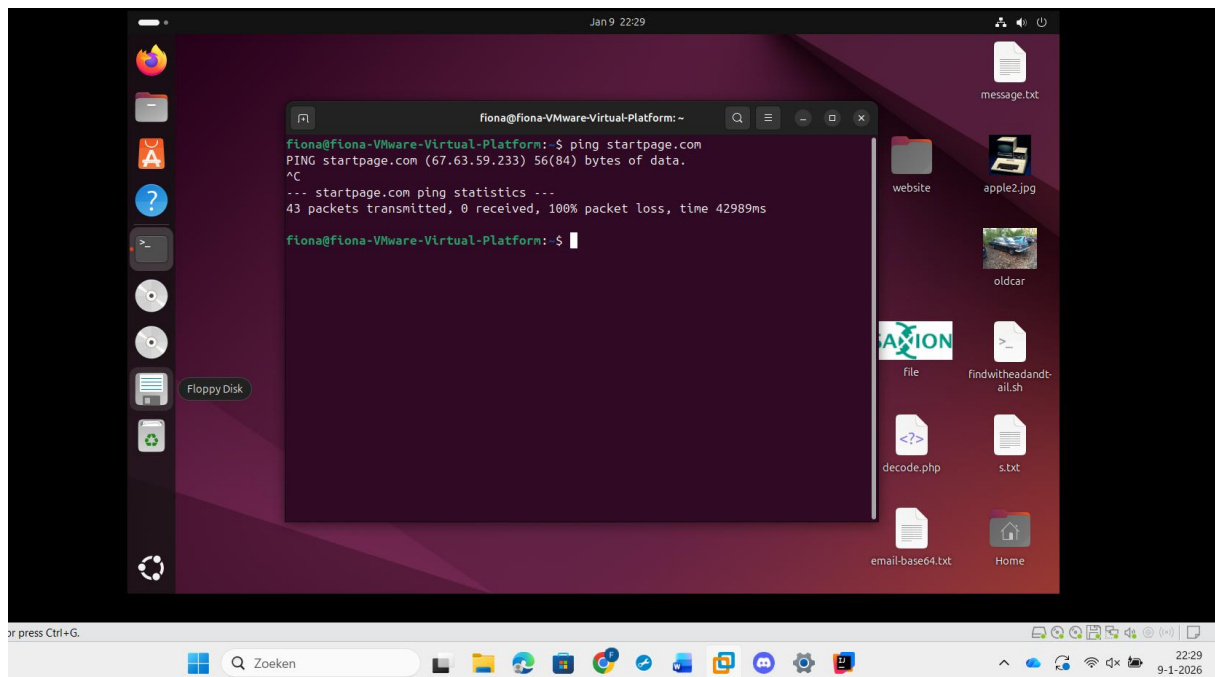
Template Week 4 – Software

Student number: 583809

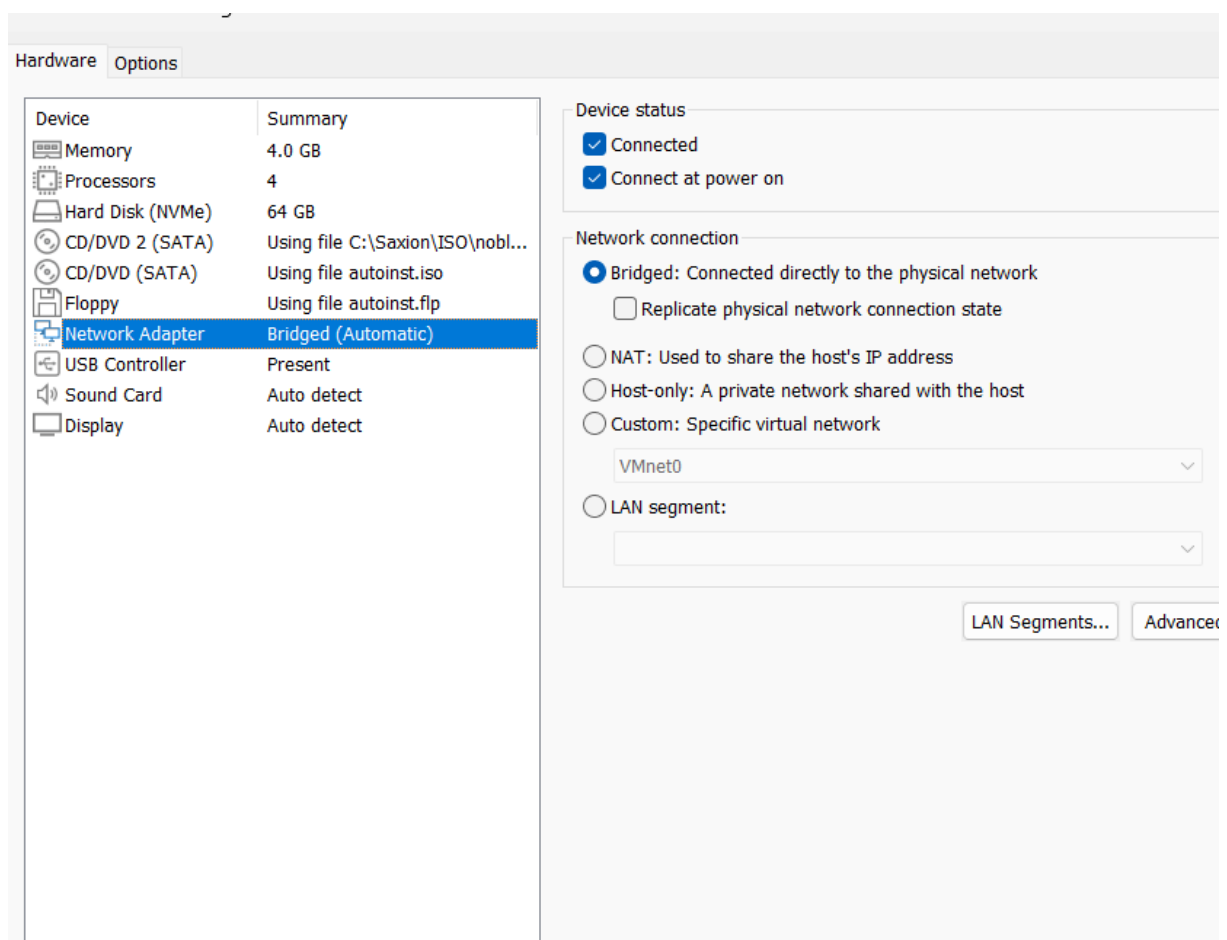
Assignment 4.1: ARM assembly

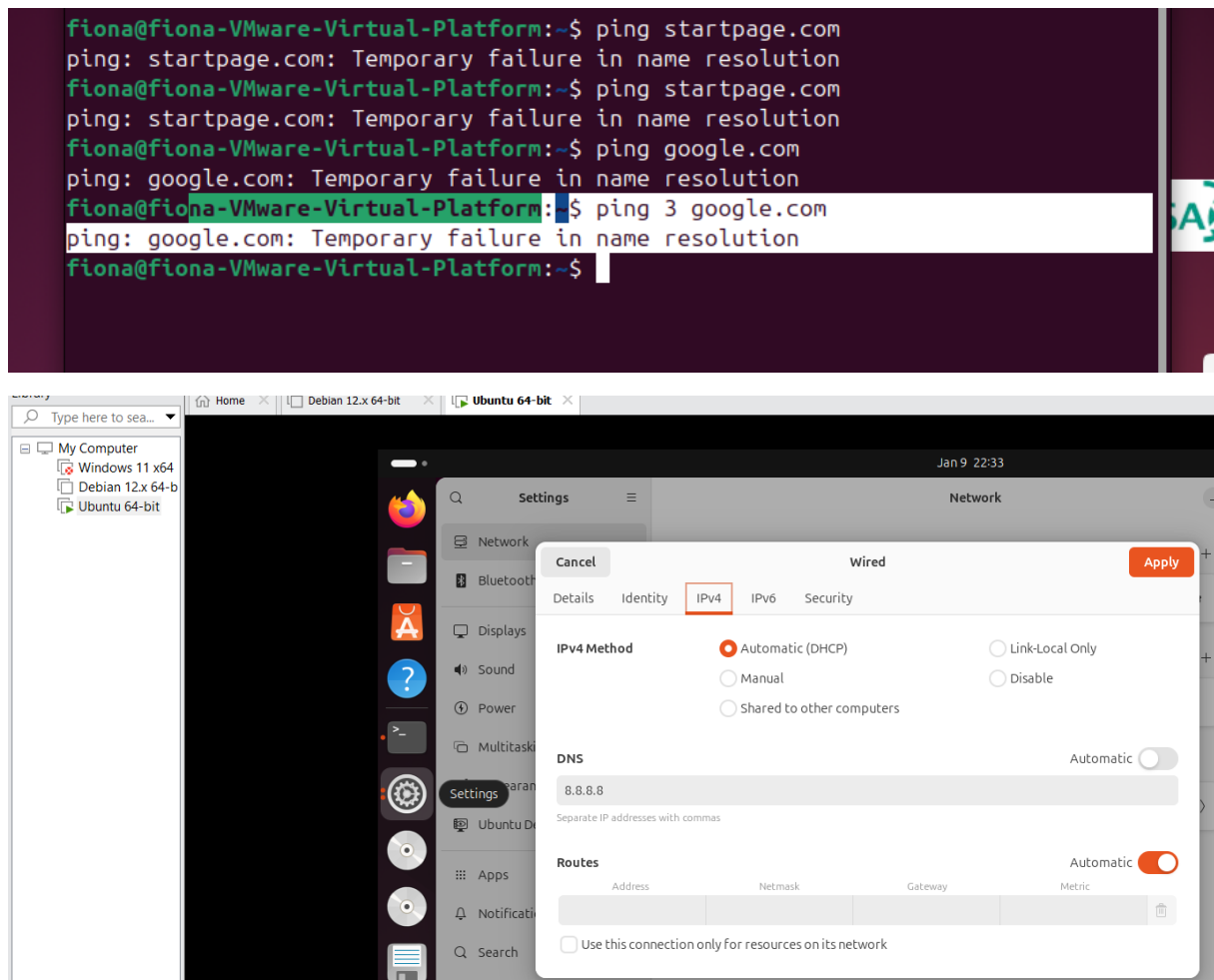
Screenshot of working assembly code of factorial calculation:





Had ineens geen internet meer bij unbunte, weet ook niet hoe ik het kon oplossen





192.168.139.3

Heb opgezocht maar kwam er niet uit, er werd aangeraden om 8.8.8.8 te gebruiken omdat die van google is.

Assignment 4.2: Programming languages

Take screenshots that the following commands work:

`javac -version`

heb ineens geen internet bij mij vm

`java -version`

heb geen internet ineens bij mijn vm

`gcc -version`

heb geen internet ineens bij mij vm

python3 --version

```
ping: google.com: Temporary failure in name resolution
fiona@fiona-VMware-Virtual-Platform:~$ python3 --version
Python 3.12.3
fiona@fiona-VMware-Virtual-Platform:~$
```

bash --version

```
Python 3.12.3
fiona@fiona-VMware-Virtual-Platform:~$ bash --version
GNU bash, version 5.2.21(1)-release (x86_64-pc-linux-gnu)
Copyright (C) 2022 Free Software Foundation, Inc.
License GPLv3+: GNU GPL version 3 or later <http://gnu.org/licenses/gpl.html>

This is free software; you are free to change and redistribute it.
There is NO WARRANTY, to the extent permitted by law.
fiona@fiona-VMware-Virtual-Platform:~$
```

Assignment 4.3: Compile

Which of the above files need to be compiled before you can run them?

Which source code files are compiled into machine code and then directly executable by a processor?

Which source code files are compiled to byte code?

Which source code files are interpreted by an interpreter?

These source code files will perform the same calculation after compilation/interpretation. Which one is expected to do the calculation the fastest?

How do I run a Java program?

How do I run a Python program?

How do I run a C program?

How do I run a Bash script?

If I compile the above source code, will a new file be created? If so, which file?

Take relevant screenshots of the following commands:

- Compile the source files where necessary
- Make them executable
- Run them
- Which (compiled) source code file performs the calculation the fastest?

Assignment 4.4: Optimize

Take relevant screenshots of the following commands:

- a) Figure out which parameters you need to pass to **the gcc** compiler so that the compiler performs a number of optimizations that will ensure that the compiled source code will run faster. **Tip!** The parameters are usually a letter followed by a number. Also read **page 191** of your book, but find a better optimization in the man pages. Please note that Linux is case sensitive.
- b) Compile **fib.c** again with the optimization parameters
- c) Run the newly compiled program. Is it true that it now performs the calculation faster?
- d) Edit the file **runall.sh**, so you can perform all four calculations in a row using this Bash script. So the (compiled/interpreted) C, Java, Python and Bash versions of Fibonacci one after the other.

Assignment 4.5: More ARM Assembly

Like the factorial example, you can also implement the calculation of a power of 2 in assembly. For example you want to calculate $2^4 = 16$. Use iteration to calculate the result. Store the result in r0.

Main:

```
mov r1, #2
    mov r2, #4
    mov r0, #1
```

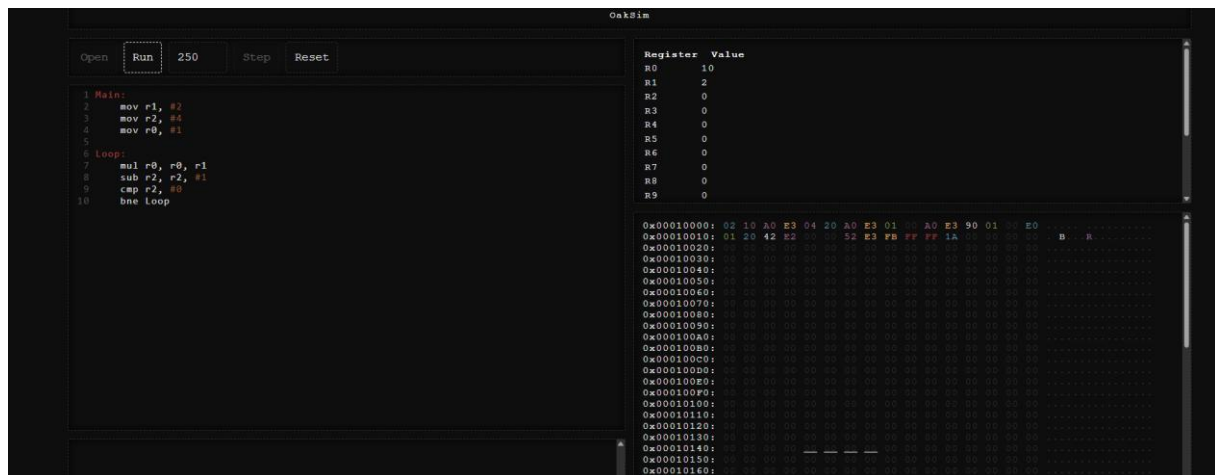
Loop:

```
mul r0, r0, r1
    sub r2, r2, #1
    cmp r2, #0
    bne Loop
```

End:

Complete the code. See the PowerPoint slides of week 4.

Screenshot of the completed code here.



Ready? Save this file and export it as a pdf file with the name: [week4.pdf](#)